A Basis for the Dyadic Sections of Euler's Totient

A structure theorem, a sharp Farey denominator bound, and the exact boundary of a kernel-checked attack on Erdős #249

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Abstract

Erdős #249 asks whether $S = \sum_{n\geq 1} \varphi(n)/2^n$ is irrational. It is open, nothing here closes it, and this note reports what a kernel-checked attack does yield, each claim bounded by the exact proposition a proof checker accepted — the negative results included.

For $j \geq 0$ and $0 \leq r < 2^j$ let $\varphi_{j,r} \colon n \mapsto \varphi(2^j n + r)$. Main theorem: $\varphi_{0,0}$, $\varphi_{1,0}$ and the sections $\varphi_{j,r}$ with r odd form a $\mathbb{Q}$-basis of the span of *all* the $\varphi_{j,r}$, so that span is infinite-dimensional, its level-e part has dimension exactly $2^e + 1$, and the only $\mathbb{Q}$-linear relations among dyadic sections of φ are those generated by two elementary identities.

Second, $S \neq a/q$ for every $a \in \mathbb{Z}$ and every $q \leq 79\,639\,646\,646\,701\,375\,323\,355\,774\,875\,831\,053$, a Farey exclusion whose method is standard: we name it first, claim no novelty, and add only that the constant is *exactly optimal* for its window. Third, the Lambert weight $A = \varphi * \mu$ of S is unbounded and no fixed positive integer clears its coordinates $A(n)/n$; that is what separates #249 from three settled neighbours. Every equivalence below is labelled one: an equivalence is not progress.

Status. The problem treated here is open, and this note does not close it. Every statement below marked as checked is a proposition that the pinned Lean kernel accepts from the sources this note links to, with no `sorry`, no added axiom, and no unchecked evaluation. That is a claim about the formal statement, not about its mathematical interest, its novelty, or the original problem. The unresolved obligations are named exactly, in their own section, and none of the finite computations, reductions, or no-go results here removes one of them.

1 Results

Every item is a proposition the pinned Lean kernel accepted, or (R9) the problem itself; the tag gives its status in the claim registry's taxonomy, and each *Checked* line links the declarations at commit `571ec44f2aad`. Nothing in the list is an irrationality criterion for S, and no combination of the items becomes one. Throughout $\varphi_{j,r}(n) = \varphi(2^j n + r)$, for $j \geq 0$ and $0 \leq r < 2^j$, is the (j, r) *dyadic section* of Euler's totient, a vector in $\mathbb{Q}^{\mathbb{N}}$. Two elementary identities relate them: the *reductions* $\varphi_{j+1,0} = 2^j \varphi_{1,0}$ (checked) and $\varphi_{j,2^{t+1}s} = 2^t \varphi_{j-t-1,s}$ for s odd with $2^{t+1}s < 2^j$ (checked).

Authorship and AI use. The prose in this version was generated by large-language-model agents under Will Cook's direction. Cook set the objectives and acceptance criteria, reviewed and authorised the manuscript, and is responsible for its contents. The agents are production tools, not authors. See *Statements and declarations*.

R1. [PROVED HERE] **Basis theorem.** The family $\mathcal{B} = \{\varphi_{0,0}, \varphi_{1,0}\} \cup \{\varphi_{j,r} : j \geq 1,\ r \text{ odd},\ 0 < r < 2^j\}$ is linearly independent over $\mathbb{Q}$ and spans the same $\mathbb{Q}$-subspace of $\mathbb{Q}^{\mathbb{N}}$ as the set of *all* dyadic sections $\{\varphi_{j,r} : j \geq 0,\ 0 \leq r < 2^j\}$; so $\mathcal{B}$ is a basis of that span, which is therefore infinite-dimensional. Since the reductions make every remaining section a rational multiple of a member of $\mathcal{B}$, the $\mathbb{Q}$-linear relations among the dyadic sections of φ are exactly those the reductions generate — a one-line consequence of the checked statements, not a further one. *Checked:* independence, span equality, basis.

R2. [PROVED HERE] **Exact finite-level rank.** For every $e \geq 0$ the $2^e + 1$ sections $\varphi_{0,0}$, $\varphi_{1,0}$ and $\varphi_{j,r}$ with $1 \leq j \leq e$, r odd, $0 < r < 2^j$ are linearly independent over $\mathbb{Q}$, so their span has dimension exactly $2^e + 1 = 2 + \sum_{j=1}^{e} 2^{j-1}$: exact, not an estimate, because the even residues are already dependent by the reductions. R1 follows by finite character of linear independence. *Checked:* independence, dimension.

Taxonomy adjudication for R1–R2. The registry groups these declarations with broader certificate and infinite-dimensional consequences under one coarse *unconditional progress* claim. The two rows above deliberately use *proved here*: for the narrower basis and exact-rank propositions this records kernel acceptance and makes no novelty assertion. Promoting either split row would require a separate prior-art judgement, not merely a declaration link.

R3. [FORMALISED HERE] **Denominator exclusion, sharp for its window.** If $S = a/q$ with $a \in \mathbb{Z}$ and $q \geq 1$ then $q > 79\,639\,646\,646\,701\,375\,323\,355\,774\,875\,831\,053 \approx 7.96 \times 10^{34}$. The constant is optimal for the window producing it: the next integer up is the exact first denominator at which that window's certificate fails. The method is Farey (Section 4). Here *formalised here* is the registry's evidence-status label: known mathematics represented and checked in Lean, not a novelty claim; the method is classical. *Checked:* exclusion, reduced-denominator form, first failure.

R4. [PROVED HERE] **Carry anti-compression.** If S were rational then some tempered integral carry orbit for the coefficients φ, with positive multiplier, would have for *every* e carry sections through level e spanning a space of dimension at least $2^e - 1$. *Closes:* every rationality proof that would exhibit a carry of bounded section rank. *Does not close:* anything else — it is not an irrationality criterion, because no finite-rank upper bound is proved for the carries that rationality supplies. The proposition is unconditional in its hypotheses, but its proof is routine transport from R2 and has no separate prior-art assessment; *proved here* therefore records kernel acceptance without making the stronger contribution judgement *unconditional progress*. *Checked:* the barrier, rank transport from R2.

R5. [FORMALISED HERE] **The #249 weight.** $S = \sum_{d \geq 1} A(d)/(2^d - 1)$ with $A = \varphi * \mu$; here $A \geq 0$, $A(p) = p - 2$, $A(p^k) = \varphi(p^k) - \varphi(p^{k-1})$, and A is unbounded. *Checked:* identity, sign, primes, prime powers, unboundedness.

R6. [FORMALISED HERE] **Eventually periodic weights are settled.** For every integer base $b \geq 2$, if $\gamma\colon \mathbb{N} \to \mathbb{Q}$ is eventually periodic, nonnegative and positive at some positive index of its periodic part, then $\sum_a \gamma(a)/(b^a - 1)$ is irrational (Luca and Tachiya prove a broader periodic theorem by other methods [3]; no priority is claimed). By R5 the weight A is unbounded, hence not eventually periodic, so #249 lies just outside this class. *Checked:* the periodic theorem.

R7. [PROVED HERE] **No fixed index clears the normalised weight.** For every integer $D \geq 1$ some $n \geq 1$ has $D \cdot A(n)/n \notin \mathbb{Z}$, and any D clearing every coordinate up to a horizon $N \geq 4$ is divisible by an explicit two-tier primorial (4 at $p = 2$, p^2 when $p^2 \leq N$, else p). *Closes:* every argument clearing the coordinates $A(n)/n$ by a single positive integer valid at all n.

Does not close: arguments whose index grows with n, or which never clear the coordinates. *Checked:* no fixed index, the primorial divisibility.

R8. [CONDITIONAL REDUCTION] **Two exact reformulations, offered as nothing more.** Irrationality of S is equivalent to a pointwise finite-certificate condition, and follows from a supply of gap certificates — one window (N, K) per denominator q whose totient carry residue avoids a band of width $q(N+K+2)$ out of 2^K. Neither hypothesis is proved for one unbounded family. Both are restatements of #249 in other coordinates, and are the part of this work most likely to be mistaken for a result. The mixed tag is intentional: it classifies the open-target content of the row. Lean proves the exact equivalence and completeness statements, but the required unbounded certificate supply remains open; proved exactness does not make the reduction unconditional. *Checked:* the equivalence, the supply implication.

R9. [OPEN] **Erdős #249.** Whether S is irrational. Not proved here.

2 Where #249 sits

If $f = g * \mathbf{1}$, that is $f(n) = \sum_{d|n} g(d)$, then interchanging the two sums gives

$$\sum_{n\geq 1} \frac{f(n)}{2^n} = \sum_{d\geq 1} \frac{g(d)}{2^d - 1}. \tag{1}$$

Four classical coefficient sequences have this shape. The table records the constant, the weight $g = f * \mu$, and its status. Decimals were recomputed to 50 places for this note and are truncated, not rounded.

f	$\sum f(n)/2^n$	weight $g = f * \mu$	status
τ	$1.60669515\ldots = E$	$g \equiv 1$	Irrational (Erdős 1948); formalised here in the Lambert form $\sum_{d\geq 1}(2^d-1)^{-1}$ (checked), equal to $\sum \tau(n)/2^n$ by (1)
ω	$0.51694281\ldots$	$g = \mathbf{1}_{\text{primes}}$	Irrational (Tao–Teräväinen 2025) [1]
Ω	$0.58950329\ldots$	$g = \mathbf{1}_{\text{prime powers}}$	Irrational (*ibid.*, remark)
φ	$1.36763080\ldots = S$	$g = A = \varphi * \mu$	**Open** (Erdős #249)

The first three weights are bounded, taking only the values 0 and 1 (for τ the weight is the constant 1, so only the value 1). The fourth is unbounded, by R5. That difference is the one that matters for the route in R6, and we do not claim it is the only difference: A also vanishes on $n \equiv 2 \pmod 4$, while $\mathbf{1}_{\text{primes}}$ is bounded but no more eventually periodic than A is, so the settled ω and Ω cases are not instances of R6 either. They were settled by a different method, discussed in Section 8.

The difficulty of #249 does not live in the Lambert transform. The neighbouring rungs are rational and exactly computable — $\sum_{d\geq 1} \mu(d)/(2^d - 1) = \frac{1}{2}$ (checked) and $\sum_{d\geq 1} \varphi(d)/(2^d - 1) = 2$ (checked) — and the signed squared-Mersenne lens $S = \frac{1}{2} + \sum_{d\geq 1} \mu(d)/(2^d - 1)^2$ (checked) recomputes S to all 50 places. The difficulty lives on the power-series side, and R1 is a statement about exactly that side.

3 The basis theorem

The producer for R2 is a separated evaluation minor. For each level, the Chinese remainder theorem together with Dirichlet's theorem on primes in arithmetic progressions supplies a set of evaluation points at which the section matrix has a nonzero minor with the required two-adic separation: one affine channel is made prime, and every other channel receives a fresh prime divisor congruent to 1 modulo a large power of two, so the columns carry distinct exact powers of two and the determinant cannot vanish (checked). Even residues never enter, because the second reduction has already carried them to odd-core channels (checked).

R1 then needs only two further steps, and both are short. Linear independence is a property of finite subfamilies, so the level-e producer, applied with e above the largest level occurring in a given finite subset, already gives independence of the whole infinite family $\mathcal{B}$. And the two reductions carry every remaining section onto a rational multiple of a member of $\mathcal{B}$, so the spans agree. That the assembly is short is the point: the level-e theorem was always the whole content, and stating its consequence as infinite-dimensionality understated it. The sharp form is a basis.

Remark. R1 and R2 are statements about φ, not about S; they are true, and their proofs are complete, whether or not S is irrational. They are here because they are the exact form of "no finite amount of the totient's dyadic structure determines the rest" — the property that R6 consumes for an eventually periodic weight and that A lacks.

The rationality side. Binary long division turns a hypothetical rational value into an integer carry recurrence: for coefficients $c(n) \le n$, and in particular for $c = \varphi$, the series $\sum c(n)/2^n$ is rational exactly when a tempered integer carry orbit exists (checked). Transporting R2 through that recurrence gives R4. The scope line there is worth repeating: this is an anti-compression barrier, and the complementary finite-rank upper bound, which is what would make it a proof, is not available.

4 The denominator bound

R3 is a Farey exclusion, and we say so before saying anything else about it. The argument is the classical mediant one. The Lean lemma commits the first 240 binary digits of the shifted series as a single residue over the totient carry window $(N, K) = (1, 240)$; a rational a/q can equal S only if it falls into a resulting bad interval of width $243/2^{240}$; that interval is bracketed by two explicit unimodular fractions $a_1/b < c_1/d$ with $bc_1 - a_1 d = 1$; and the mediant lemma (checked) forces any rational strictly between unimodular neighbours to have denominator at least $b + d$. The excluded range is therefore $q \le b + d - 1$, which is the printed constant (checked), and the transport to the series is a tail estimate (checked). *The method is standard, and we claim no novelty for it.*

Two things should be read off correctly. First, the scale. A window of K computed binary digits yields a mediant bound of order $2^{K/2}$: the same pipeline gives 2.49×10^{17} at $K = 120$ (checked) and 7.96×10^{34} at $K = 240$. The constant is a function of how many digits were committed, not a measure of how much the problem has moved. Second, the sharpness. Within its window the bound cannot be improved at all: $b + d$ itself fails the certificate, so $b + d - 1$ is exactly the last denominator excluded (R3). A reader who regards an explicit Farey exclusion as routine is not in disagreement with this note. R1, not R3, is where we would ask a sceptical reader to look.

The bound transfers to the coprimality form of the constant with the constant halved. Writing the visible-coprime-pair series as $\sum_{\gcd(m,n)=1,\ m,n\geq 1} 2^{-(m+n)}$, one has $S = \frac{1}{2} + \sum_{\gcd(m,n)=1} 2^{-(m+n)}$ (checked), so that series has no representation a/d with $d \leq 39\,819\,823\,323\,350\,687\,661\,677\,887\,437\,915\,526$ (checked), that number being $(q_0 - 1)/2$ for the constant q_0 of R3. This series is the probability that two independent geometric fair-coin waiting times are coprime; the checked statements are the series identity and the series exclusion, and the probabilistic reading is a gloss on them rather than a formalised measure-theoretic statement.

5 Barriers, and two we do not state as theorems

R7 is the one method barrier this note states, and its scope lines are given with it in Section 1. It is worth a sentence of mechanism, because the mechanism is the whole of it: take a prime $p > \max(D, 2)$; then $A(p) = p - 2$ by R5, and $p \nmid D(p-2)$, so $D \cdot A(p)/p \notin \mathbb{Z}$. Being one line is a virtue here. The statement is about the weight the abstract already names, it needs no vocabulary this note has not defined, and it survives a hostile reading.

Two further no-go results in the formal corpus appeared as propositions in an earlier version of this note, and are demoted rather than dropped, because the reason is more useful to a reader than the propositions were.

A fixed-precision transport result (checked) says that at every fixed positive precision, each finite compatible valuation-unit word admits a prefix-locked centred completion. Its statement quantifies over abstract pairs of a two-adic valuation and an odd unit; it mentions neither φ nor S, its content is that a congruence class meets any interval of the corresponding length, and it is a restatement of a lemma printed immediately above it. The class of arguments it closes — deriving a contradiction from a local signature read at a precision fixed in advance, along a finite word — is not one anybody has proposed for #249.

A factor-ideal result (checked, with a sparse-anchor companion at checked) exhibits, for each $t \geq 3$ with $H = \mathrm{lcm}(1, \ldots, t)$, a nonzero integer carry whose forcing letters lie in the ideal generated by $\varphi(H)$, which reproduces the true totient differences at $t - 2$ prescribed indices, and which every finite integer shift polynomial carries to a pair with the same coboundary form, the same ideal memberships and an ℓ^1-weight bound. It does close that hypothesis class. Two limits. The witness is a spike — its state is $-\varphi(H)$ at $t - 2$ points and zero elsewhere, so it stays uniformly bounded while the strip bounds it respects grow, and the construction is cheap for that reason. And the shifted pairs are not asserted to be nonzero, so an argument that additionally demanded non-vanishing after the shift is not closed. An earlier draft of this note also attributed to it a whole-ray anchor condition, a diagonal-bound condition and a strict-survivor condition that the cited theorem does not contain, and described it as "surviving" the shift algebra when the checked conclusion carries no non-vanishing clause. Correcting that is the reason it is here at all.

6 Three problem-owned programmes

The modules below are problem-owned rather than part of the reviewed corpus. Each states its unproved hypothesis explicitly; none advances #249, and we list them because a reader deciding whether to spend time here should see the sufficient conditions in the exact form Lean checks them.

Strict prime-orbit escape. The cleanest of the three, because it moves a threshold. The producer asks for cofinally many primes p at which the first tail-orbit exponential has real part strictly below 9/10. Granting it, a positive adaptive truncation budget follows, and then S is irrational. The point of interest is the constant: the legacy route required a 4/5 gap, this one requires only 9/10, so the sufficient condition is strictly weaker while the conclusion is unchanged. Weakening a hypothesis you still cannot verify is progress on the *statement*, not on the problem, and the producer remains unproved.

Finite Euler-sieve algebra. Elementary identities behind the squared-Möbius approximation to the totient generating function; no transcendence input is claimed. The local Euler factors are exact squares at $s = 1$ and $s = 2$, and on the divisor-sum prime-power row the first difference is $p-1$ while the second difference is $p^{e+1}(p-1)$. The second difference is the content: applying $\mu * \mu$ converts the divisor-sum row into the prime-power totient row exactly, at every finite stage. These are one-line facts; they are recorded because the approximation they license is otherwise easy to assert without checking.

Prime-ray cyclotomic curvature. A producer/consumer interface with two named obligations. Its curvature functional is the checkerboard on a two-anchor rectangle, which annihilates every separable background $u(x)+v(y)$ and is the unique such 2×2 annihilator up to scale. That uniqueness is worth stating: the operator is not a convenient choice among many, it is the only one, which is what makes the resulting obstruction coordinate-free. On top of it sit a squarefree divisor-layer cancellation and a centred-completion step for residue classes modulo an even modulus. The two obligations are eventual clean cyclotomic layers on a prime ray and an exact-order witness at every rational prime divisor of a layer; together they give escape from every prescribed finite prime support. Neither obligation is proved here, and polynomial resultant realisability and Archimedean growth remain upstream of both.

7 What we need from a mathematician

Three obligations carry the whole conditional part of this work. Each is a statement about one explicitly defined sequence, each is stated below from definitions given here, and none needs the rest of this note — or any Lean — to be understood, proved or refuted. They are written in the form in which we would want an answer.

Two remarks apply to all three. A negative answer is worth exactly as much to us as a positive one: if a route provably cannot supply what it is asked for, we will say so here and drop the route rather than leave it listed as live. And none of the three asks anyone to grant anything about S; each is an ordinary question about a defined object, answerable, refutable or recognisable from the literature on its own terms.

Ask 1. Certificates beyond every cutoff, on the lcm **diagonal.** Write $H_t = \operatorname{lcm}(1, \dots, t)$ (definition) and let $R_N = \sum_{m \ge 1} \varphi(N+m)/2^m$ be the fractional layer of $2^N S$ (definition). *Wanted:* for every t_0 there is a $t \ge t_0$ with $R_{2H_t} - R_{H_t} \notin \mathbb{Z}$. That is the hypothesis on the right of the diagonal normal form, in the finite-certificate coordinates the two forms share (checked).

Already settled, so that nobody repeats it. The normal form is an equivalence, not a sufficient condition: S is irrational if and only if that supply exists. Nothing short of a cofinal statement is therefore of use, and no finite band approximates one. The converse direction is sharper than it looks — one pair $h > 0, N$ with $R_{N+h} - R_N \in \mathbb{Z}$ already makes S rational (checked). The certificate is decidable at each t, and has been kernel-checked to fire at the 28 scales (listed) lying between $t = 1$ and $t = 64$ (checked). Lengthening that list is not progress towards the ask, and is not offered as such: a band bounded at $t = 64$ contributes nothing at $t = 65$.

Most useful first. (i) Any argument giving $R_{2H_t} - R_{H_t} \notin \mathbb{Z}$ for infinitely many t, with no uniformity or rate required. (ii) A reason the lcm diagonal is the wrong ray to ask this of — for instance that $R_{2H_t} - R_{H_t}$ is beyond the reach of present methods for structural reasons, not merely unproved. (iii) A single t with $R_{2H_t} - R_{H_t} \in \mathbb{Z}$. A construction or a proof (i) helps most; (ii) is next and would be acted on.

If answered. (i) closes #249, since the normal form is an equivalence and the deduction is already checked. (ii) retires the diagonal presentation, and with it Ask 1, without touching the problem. (iii) settles #249 in the opposite direction, which would be a larger result than anything asked for here.

Ask 2. One cofinal prime with first-harmonic real part below $9/10$**.** *Wanted:* for every $h \ge 1$ and every N_0, a prime $p \ge \max(N_0 + h + 1, h + 5)$ with

$$\cos(2\pi\,(R_{p-1} - R_{p-h-1})) < \tfrac{9}{10},$$

equivalently with $R_{p-1} - R_{p-h-1}$ further than $\arccos(9/10)/2\pi = 0.07178\ldots$ from the nearest integer. Written with $N = p - h - 1$, this is the hypothesis verbatim: the real part of the first additive character of $R_{N+h} - R_N$, below $9/10$ for cofinally many primes.

Already settled. The quantity is not an artefact of the truncation: integer prefixes cancel under the character, so this is exactly the first additive character of $2^N(2^h - 1)S$ (checked), that is, of one orbit of the doubling map. Granting the gap, the adaptive truncation budget (checked) and then irrationality (checked) follow with no further input. The threshold has already been moved: the legacy route consumed a $4/5$ gap and this one consumes $9/10$, so the hypothesis is strictly weaker than the one it replaced. We have no proof of it at either threshold.

Most useful first. (i) A reference. This is a one-sided escape statement for a single explicit real orbit sampled along the primes, and an analytic number theorist may recognise it as standard, as a known consequence of an existing estimate, or as a known hard case; any of the three would be decisive for how we spend time. (ii) A proof or proof sketch. (iii) A reason the statement is out of reach for orbits of this kind. (i) helps most, because the cost of the wrong answer here is measured in months.

If answered. (i) or (ii) closes #249 through the checked endpoint above. (iii) closes only this branch: the gap is sufficient for irrationality and nothing here shows it necessary, so its failure would leave the problem exactly where it is. This is the ask we consider most likely to be answerable by somebody who has not read the rest of this note.

Ask 3. Bounded multiplicative order along a prime ray. Fix integers $m, d \geq 1$ and a layer sequence $C\colon \mathbb{N} \to \mathbb{N}$. Two statements are wanted, of which the second is the substantial one. (a) There is a Q_0 with $C(mq) > 1$ and $\gcd(C(mq), mq) = 1$ for every prime $q \geq Q_0$ (the supply). (b) Whenever q and p are prime and $p \mid C(mq)$, some k with $1 \leq k \leq d$ has $mq \mid p^k - 1$ (the consumer). Statement (b) is a primitive-divisor question in classical shape: it asks that every prime divisor of a layer have multiplicative order bounded by d relative to the ray index mq.

Already settled. The curvature coordinates are forced rather than chosen — the checkerboard is the only 2×2 joint annihilator up to scale (checked) — and the divisor-layer cancellation (checked) and the affine recentring step (checked) are proved from their stated hypotheses. What is *not* there should be said plainly: (a), (b) and the intended consequence finite-prime-support escape are all recorded as definitions, and the module proves neither obligation. One line, offered as prose and not as a checked statement: (b) alone already forces the escape by size, since $p \mid C(mq)$ gives $mq \leq p^d - 1$, so for a finite set S of primes no member of S divides $C(mq)$ once $q > \max_{p \in S} p^d / m$. The role of (a) is then to keep the conclusion from being vacuous, since $C(mq) = 1$ would satisfy it trivially.

Most useful first. (i) A reference: whether (b) is known, in either direction, for cyclic resultants of an integer polynomial, which is the intended source of C. (ii) A counterexample family for which (b) fails. (iii) A construction of a layer sequence satisfying both. A reference helps most, because we do not know whether we are asking something classical or something false.

If answered. This closes one branch and not the problem. Escape from every prescribed finite prime support is not irrationality of S: realising C by cyclic resultants at all, and the Archimedean growth estimate, are both upstream, both unproved, and no route from escape to #249 is claimed here. An answer tells us whether to keep the branch; it does not move #249.

Not an ask: the denominator bound. The constant 7.96×10^{34} of R3 is not an open item, and help with it is not wanted. It is the classical Farey mediant bound applied to a window with a free parameter K, it has order $2^{K/2}$, and raising it needs only more committed binary digits — a computation, not an idea. A reader whose first instinct is to improve that number would be spending effort at the one point in this note where the method is completely understood and the return is nil.

8 What remains open

The problem is open and this note does not narrow it. Precisely:

1. Irrationality of S (R9). No proof is claimed.

2. The certificate routes of R8 convert irrationality into the existence of an unbounded supply of finite integer certificates. Finite instances are checked — R3 is one, at $K = 240$ — and the unbounded supply is not. *An equivalence is not progress.*

3. The natural analytic target — a cancellation estimate for the first-harmonic block sum attached to the doubling orbit of $(2^h - 1)S$ — is unproved, and is strictly stronger than irrationality rather than equivalent to it.

A route we would now take seriously, and do not have: adapt the Gowers-uniformity and quantitative-correlation method of [1] from the bounded weight $\mathbf{1}_{\text{primes}}$ to an unbounded weight. Their argument uses the prime-dilation identity $\omega(pn) = \omega(n) + 1 - \mathbf{1}_{p|n}$, whose main increment is constant and whose error is sparse. For φ the corresponding correction is valuation-dependent, and the formal corpus makes the difficulty precise rather than impressionistic: on any prescribed finite horizon the correction *can* be suppressed, by a bounded simultaneous square-CRT base (checked), but suppression alone forces nothing, since a clean block can vanish (checked) and can equally be nonzero (checked). So the transfer is not blocked by an obvious obstruction and is not routine either. It is the first place a reader who wants to close #249 should look, and it did not exist when this development began.

Statements and declarations

This manuscript is authored exposition, not proof authority. The linked Lean snapshot is authoritative only for its exact propositions, and kernel checking establishes that a proposition was proved, not that it is interesting, novel, or sufficient. The consequence drawn in the last sentence of R1, the deduction in R6 that an unbounded weight is not eventually periodic, the mechanism sentence in Section 5, and the size argument under Ask 3 in Section 7 are one-line arguments in the prose, marked as such; the first three are drawn from checked statements, and the last is drawn from two hypotheses that are not proved. Numerical decimals quoted in Section 2 were recomputed independently for this note. Erdős Problem #249 remains open.

References

[1] T. Tao and J. Teräväinen, *Quantitative correlations and some problems on prime factors of consecutive integers*, arXiv:2512.01739 (submitted December 2025, revised April 2026). Proves irrationality of $\sum_{n\geq 1} \omega(n)/2^n$, and remarks that the method also gives $\sum_{n\geq 1} \Omega(n)/2^n$ and every integer base $b \geq 2$.

[2] K. Pratt, *The irrationality of a prime factor series under a prime tuples conjecture*, arXiv:2409.15185. The conditional predecessor of the above.

[3] F. Luca and Y. Tachiya, *Irrationality of Lambert series associated with a periodic sequence*.

[4] P. Erdős and R. L. Graham, *Old and New Problems and Results in Combinatorial Number Theory*, 1980, p. 61.

[5] T. Bloom, *Erdős Problems*, `https://www.erdosproblems.com/249`.